Exercise 01 SQL Foundations

Selecting and Sorting

1. SELECT *
   FROM employees;

ID	First_Name	Last_name	Department	Salary	hire_date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-13	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. SELECT DISTINCT Department
   FROM Employees;

Department
IT
HR
Finance
Marketing

3. SELECT first_name, last_Name, Salary
   FROM employees
   ORDER BY salary desc;

First-name	last_name	Salary
Eva	Naidoo	72 000
Alice	Green	70 000
Cathy	Zulu	65 000
David	Mokoena	50 000
Brian	Lee	45 000

4. SELECT Id, First_name, Last_name, Salary
FROM Employees
WHERE Salary >= 65 000;

ID	First_Name	Last_name	Salary
1	Alice	Green	70 000
3	Cathy	Zulu	65 000
5	Eva	Naidoo	72 000

Filtering With WHERE

5. SELECT Id, First_name, Last_name, Department
FROM Employees
WHERE Department = "IT";

ID	First_name	Last_name	Department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT Id, First_name, Last_name, Department, Salary
FROM Employees
WHERE Finance salary > 60 000;

ID	First_name	last_name	Department	Salary
3	Cathy	Zulu	Finance	65 000

7. SELECT Id, First_name, last_name, department
FROM Employees
WHERE Department = "HR" OR "Marketing";

ID	First_name	last_name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT ID, First_name, last_name, Department
FROM Employees
WHERE Department NOT "IT";

ID	First_name	last_name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. SELECT Id, First_name, last_name, Department
FROM Employees
WHERE Department IN(IT, HR, Finance);

ID	First_Name	Last_name	Department
1	Alice	Green	IT
5	Eva	Naidoo	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

10 SELECT id, First_name, last_name, Department, Salary, City
FROM Employees
WHERE Department IN 'IT' AND Salary > 65 000
AND City = "Johannesburg";

ID	First_Name	last_name	Department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg